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Experiment-1 :- Use of Sequence Data Types in Python

Aim:- To study and implement different sequence data types in Python and perform operations on them.

Theory:- Sequence data types are used to store multiple values in an ordered manner. They allow operations like indexing, slicing, iteration, and membership testing. Programs

a) Strings

```
s = "Hello Python"
print(s[0])
#Indexing
print(s[8:5])
# Slicing
print(s.lower())
# Operation
```

b) Lists

```
lst = 10, 20, 30, 40)
lst.append(50)
print(lst)
print(lst [1:3])
```

c)Arrays

```
import array as arr
a = arr.array('i', [1, 2, 3, 4])
a.append(5)
print(a)
```

d) Tuples

```
t = (1, 2, 3, 4)
print(t[0])
print(t[1:3])
```

f) Sets

```
s = {1, 2, 3, 3, 4}
print(s)
# duplicates removed
```

g) Range

```
r = range(1, 10, 2)
print(list(r))
```

Double-click (or enter) to edit

Theory Questions & Answers

1. What are sequence data types in Python?

Sequence data types store multiple values in an ordered way and allow indexing and slicing.

Types:

- String
- List
- Tuple
- Range
- Array (from module)

Indexing & Slicing Indexing: Access element using position Example: `s[0]` → "rst element"

Slicing: Access a range Example: `s[1:4]` → elements from index 1 to 3

Works similarly in strings, lists, tuples, and arrays

Syntax: `[start : end : step]`

2. Difference Between List and Tuple

Feature	List	Tuple
Mutability	Mutable	Immutable
Syntax	<code>[]</code>	<code>()</code>
Use Case	Dynamic data	Fixed data
Performance	Slower	Faster

Why prefer tuple?

Data safety (cannot be changed)

Faster execution

Used in constants (e.g., coordinates)

3. Arrays vs Lists

Feature	List	Array (from <code>array</code> module)
Data Type	Stores heterogenous data types	Stores homogenous data types
Flexibility	More flexible, can grow/shrink	Fixed type after creation
Memory	More memory overhead	More memory efficient
Operations	General-purpose operations	Optimized for numerical operations
Usage	Common for general data collections	Primarily for numeric data, e.g., when interfacing with C
Syntax	<code>[item1, item2, ...]</code>	<code>array('typecode', [item1, ...])</code>

When to use arrays?

1. When handling large numeric data
2. When memory efficiency is important

4. How Dictionaries Work Internally

- Dictionaries use hashing
- Each key is converted into a hash value
- Data is stored using key-value pairs

5. Difference Between Sets and Lists

Feature	List	Set
Order	Ordered	Unordered
Duplicates	Allows duplicates	Does not allow duplicates
Mutability	Mutable (can change elements)	Mutable (can add/remove elements)
Indexing/Slicing	Supported	Not supported
Syntax	<code>[item1, item2, ...]</code>	<code>{item1, item2, ...}</code> (or <code>set()</code>)
Use Case	Collection of items, ordered sequences	Unique collections, membership testing

Handling Duplicates

- Sets automatically remove duplicates

Practical Use

- Removing duplicates from data
- Membership testing
- Mathematical operations (union, intersection)

Conclusion:-

In this assignment, we learned about different sequence data types in Python such as strings, lists, arrays, tuples, dictionaries, sets, and range. We also performed operations like indexing and slicing, and understood their real-world applications.

